



ALOS: Acoustic Localization System Applied to Indoor Navigation of UAVs

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Agenda

- Introduction
 - Position estimation problem.
- Proposed system architecture
 - ALOS
 - Localization algorithm
 - Kalman Filter
- Results
 - physical experiment using ALOS
 - simulation trials
- Conclusion



Position estimation

- Indoor environments are GPS denied
- Multiple technologies were studied throughout the years
 - Infrared camera systems (OptiTrack, Vicon)
 - Electromagnetic signals
 - WiFi, Zigbee, Bluetooth (received power)
 - UWB (TOF)
 - Acoustic waves (TOF)

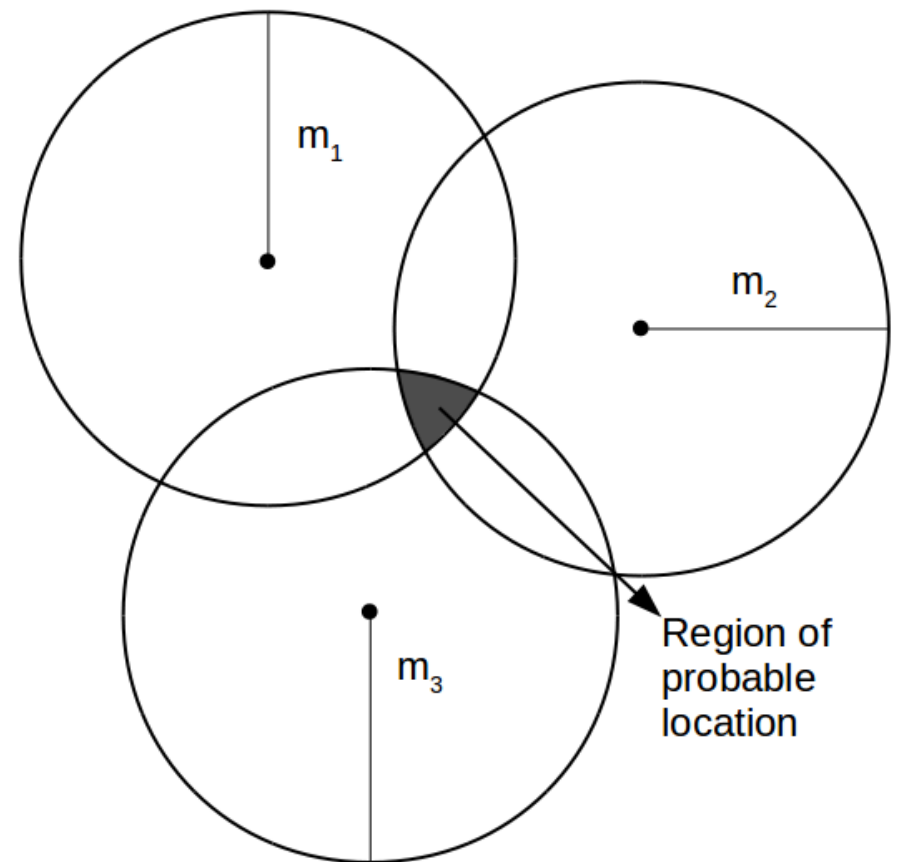


Position estimation

- In general, the accuracy in position estimation with the technologies presented is:
 - camera systems - centimeters or millimeters
 - received power electromagnetic systems - meters or UWB - centimeters
 - acoustic waves - centimeters

Position estimation using trilateration

- It is possible to measure the distance (m_i) from a source to a receiver using ultrasonic sensors
- Using multiple sources, one can calculate the position of a receiver with a trilateration algorithm

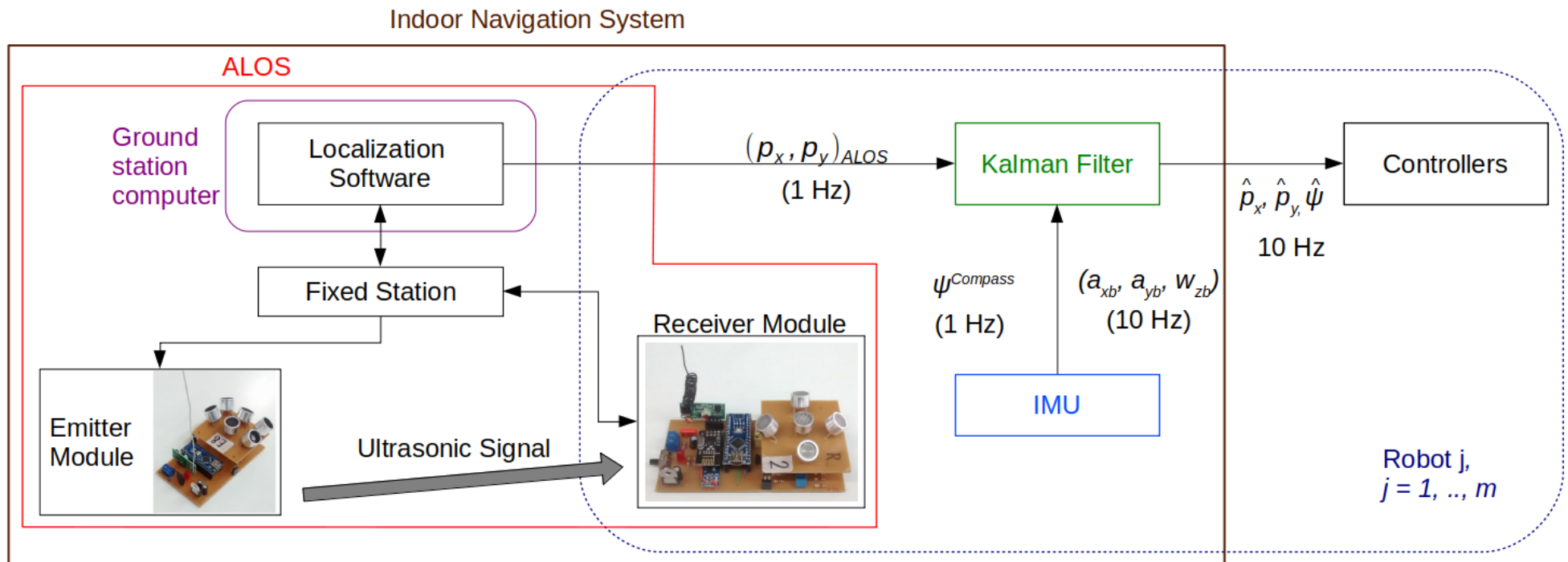




Objective

To build an ultrasonic-based indoor navigation system that is capable of compute the position and orientation of multiple aerial vehicles in real-time operation

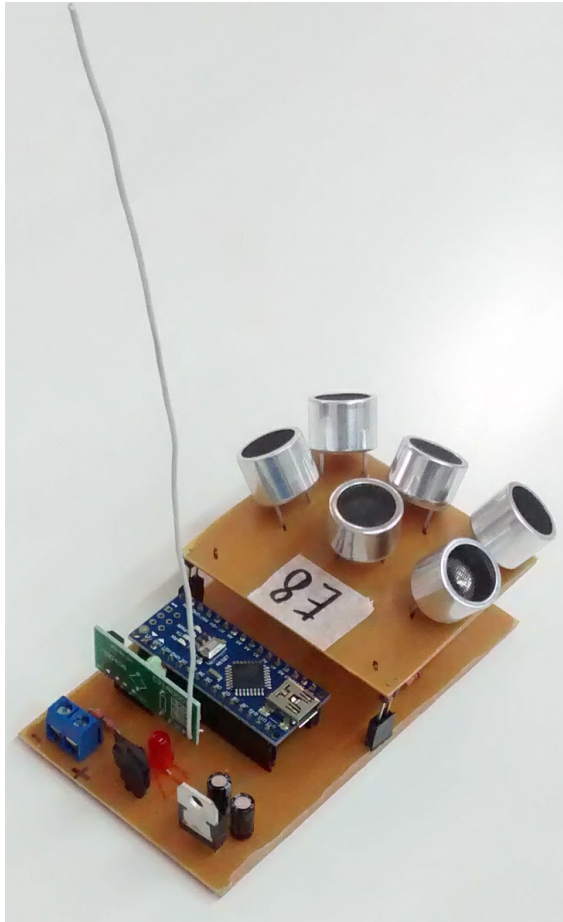
System proposed



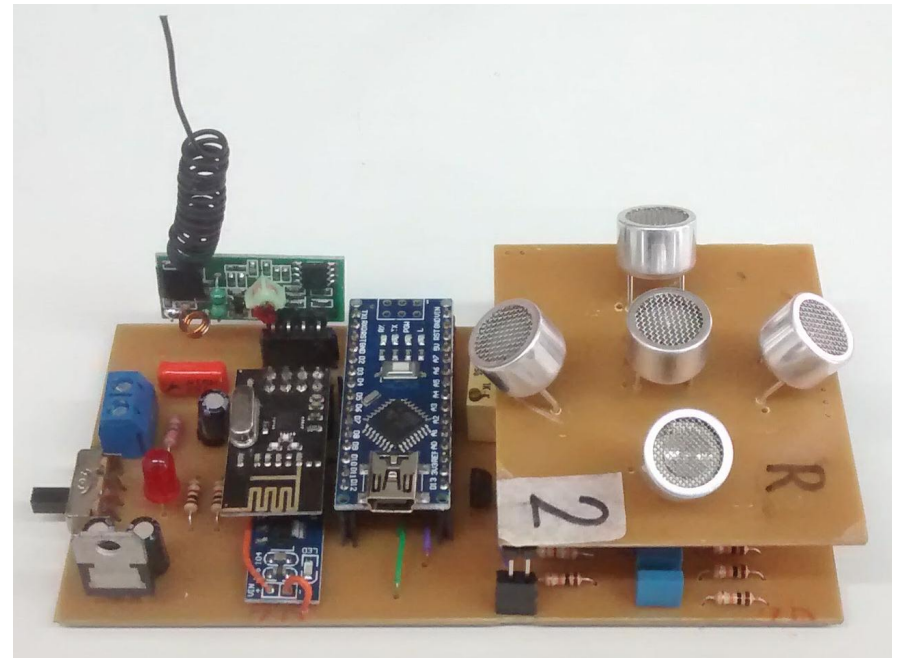
ALOS (Acoustic Localization System)

- Uses ultrasonic signals to measure the distances between fixed ultrasonic emitter modules and a ultrasonic receiver coupled to a robot
- Two radio frequency channels for communication:
 - 433 MHz - synchronization of operation
 - 2.4 GHz - data transfer
- Uses *time windows* to allocate and organize the measurements of TOF. At each *time window*, only one of the eight emitters sends ultrasonic pulses to the environment.
- Return the position of all receiver modules every second (1 Hz)

ALOS



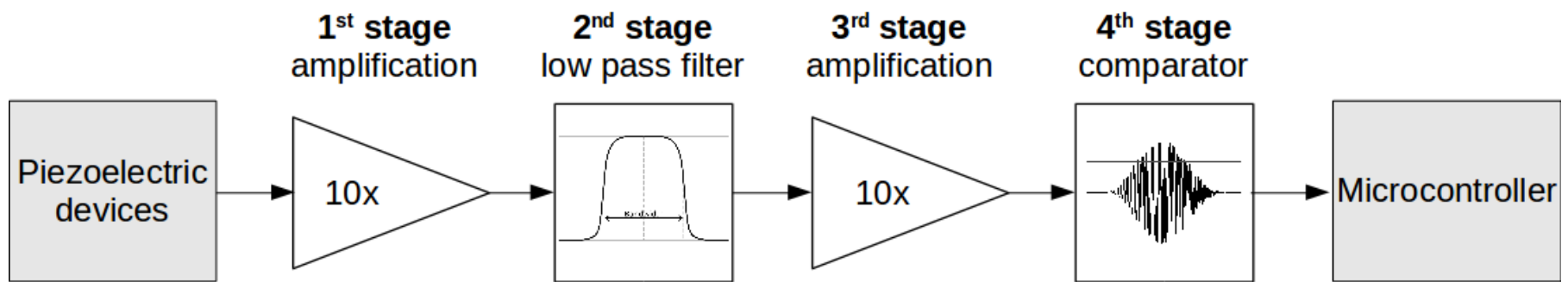
Emitter Module
(fixed)



Receiver Module
(coupled to the robot)

ALOS

- A received ultrasonic signal passes through a conditioning circuit inside Receiver Module, before being detected by the module's microcontroller.



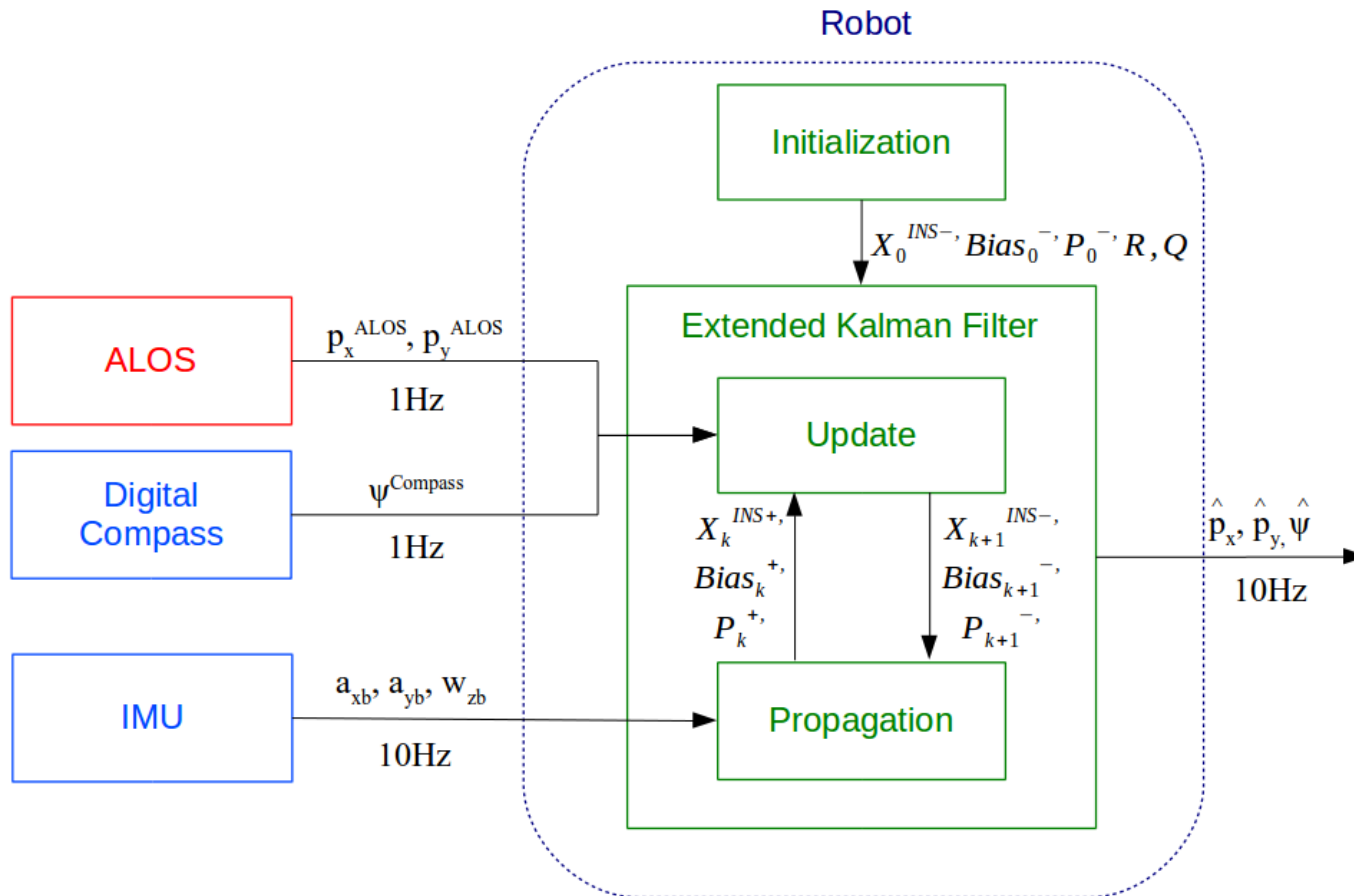
ALOS: Trilateration Algorithm

- The position of a receiver module is computed by the trilateration algorithm, that uses a Taylor Series approximation of the distances measured by each ultrasonic signal.

Algorithm 1 Calculate $p = (x, y, z)$

```
1:  $p_{initial} = (x_v, y_v, z_v)$ 
2: while  $\vec{\delta}$  is not sufficiently small do
3:   for  $i = 1, 2, \dots, n$  do
4:     /* at the point  $p_{initial} = (x_v, y_v, z_v)$  */
5:     Calculate  $f_{i,v}$ ;
6:     Calculate  $a_{i,x}, a_{i,y}$  e  $a_{i,z}$ ;
7:   end for
8:   Fill the matrix  $A$  and the vector  $\vec{D}$ ;
9:   Calculate  $\vec{\delta}$  using equation (10);
10:  Calculate  $p = (x, y, z)$  using equation (6);
11:  Make  $p_{initial} = p$ ;
12: end while
13: return  $p = (x, y, z)$ 
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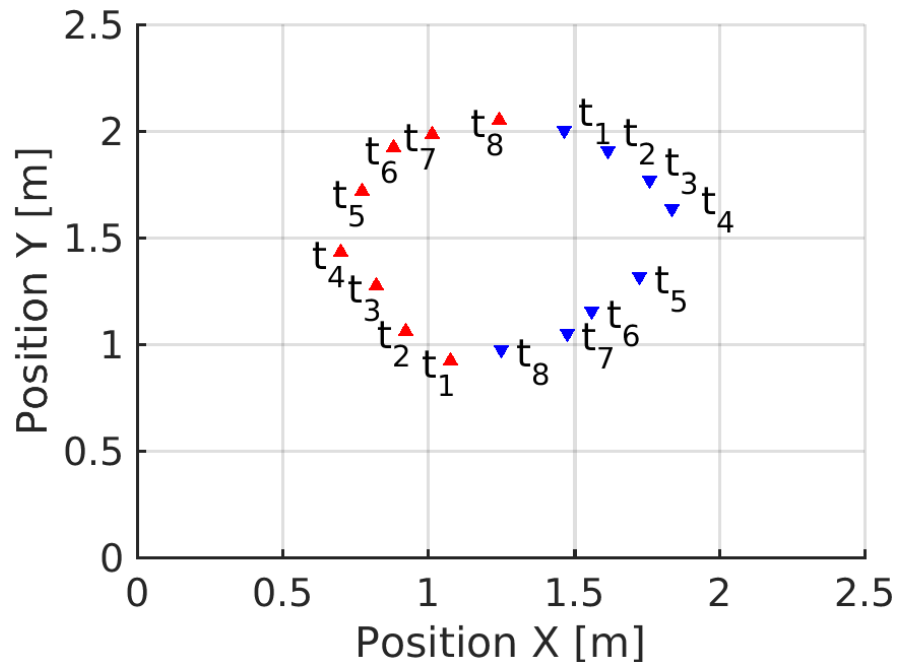
Kalman Filter - Sensor fusion and Pose estimation



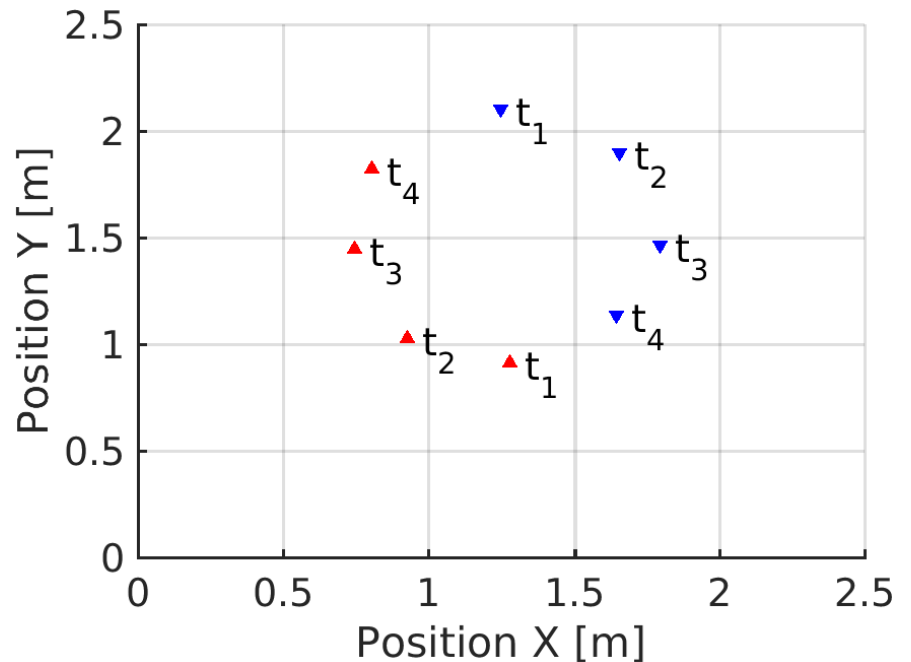
Results

physical experiment

- In this experiment, two modules were manually positioned in the environment.
- ALOS computed the position of the receiver modules every 1 s
- Accuracy in position estimation around 3 cm



Receiver Modules at ground level

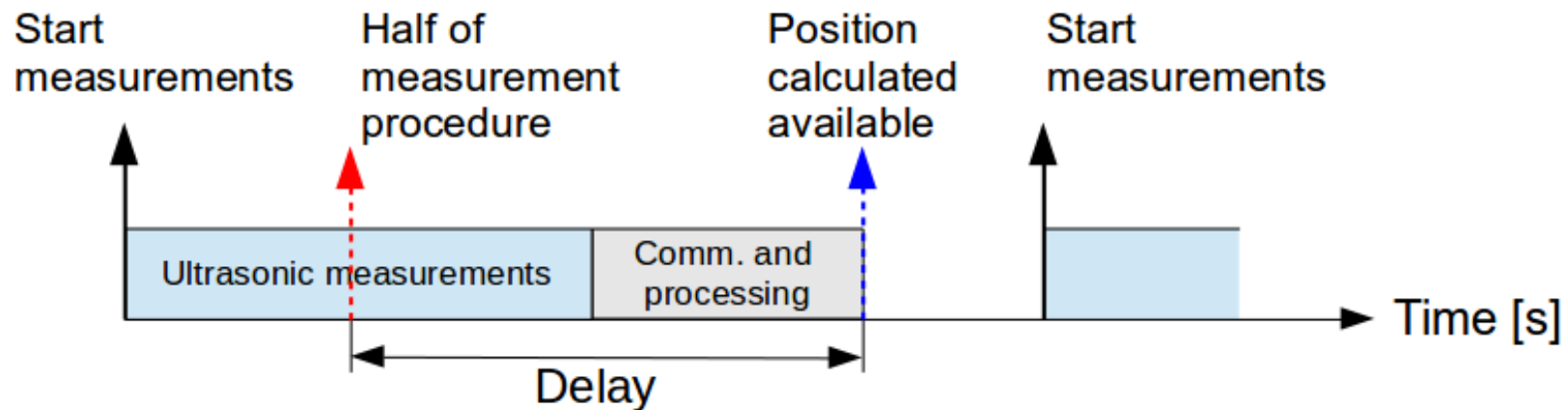


Height = 70 cm

Results

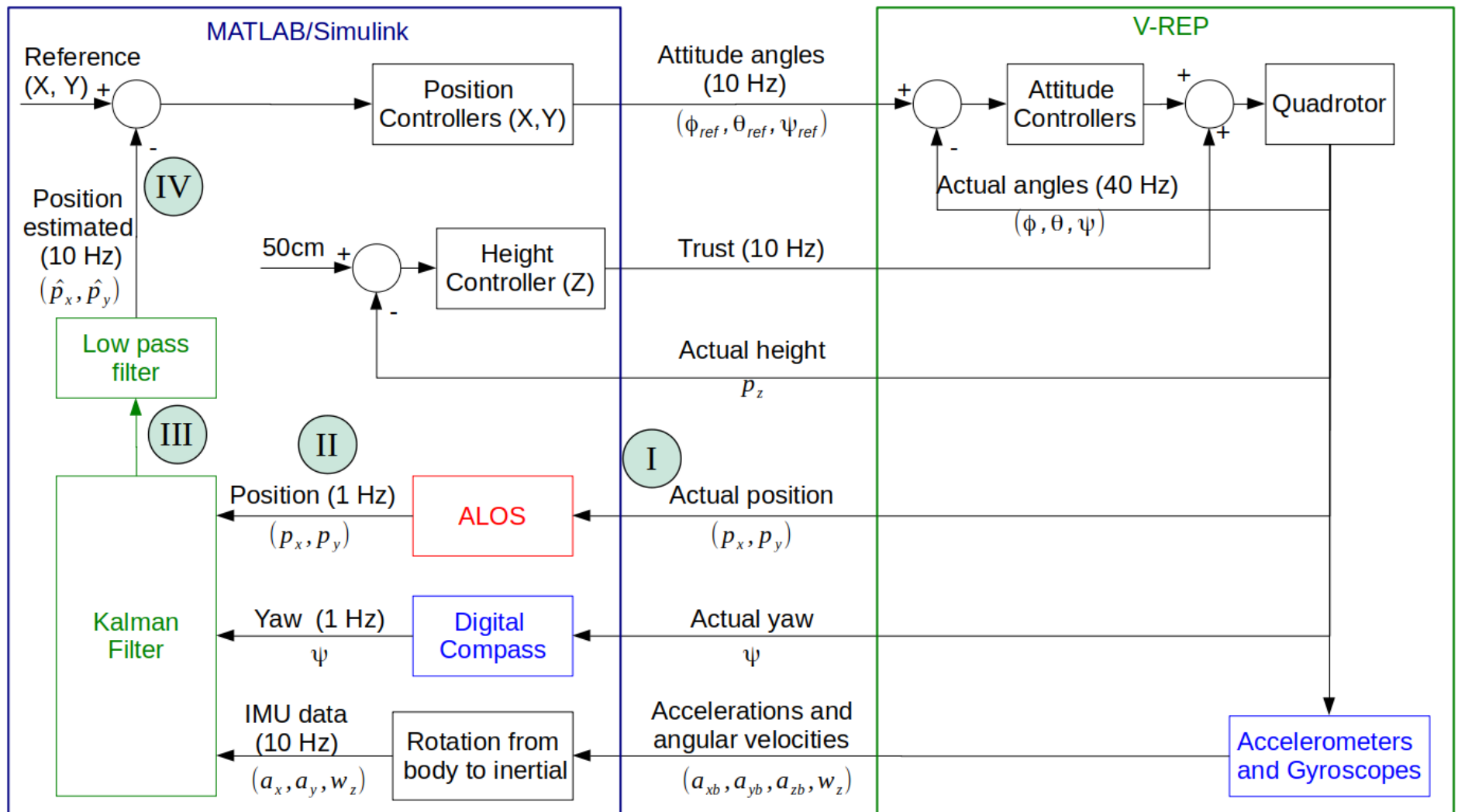
physical experiment

- We identified that ALOS has a transport delay in the position estimation.
- Consider a receiver module physically positioned in one point in space, the computed position is only available to the robot at the end of communication and processing time.



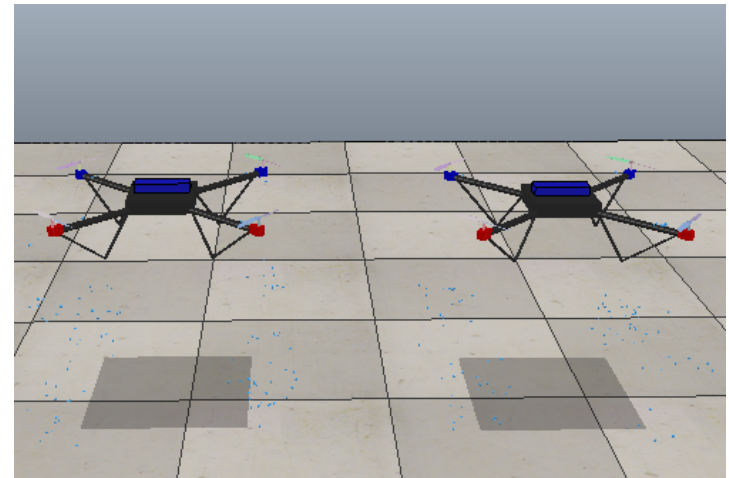
Results simulation trial

- Simulation using the complete system proposed



Results simulation trial

- Simulation environment composed of:
 - MATLAB/Simulink (position controllers, ALOS and the Kalman Filter)
 - V-Rep modules (quadrotor dynamics and attitude controllers)
 - Robot Operating System (ROS) for communication between softwares





Results simulation trial

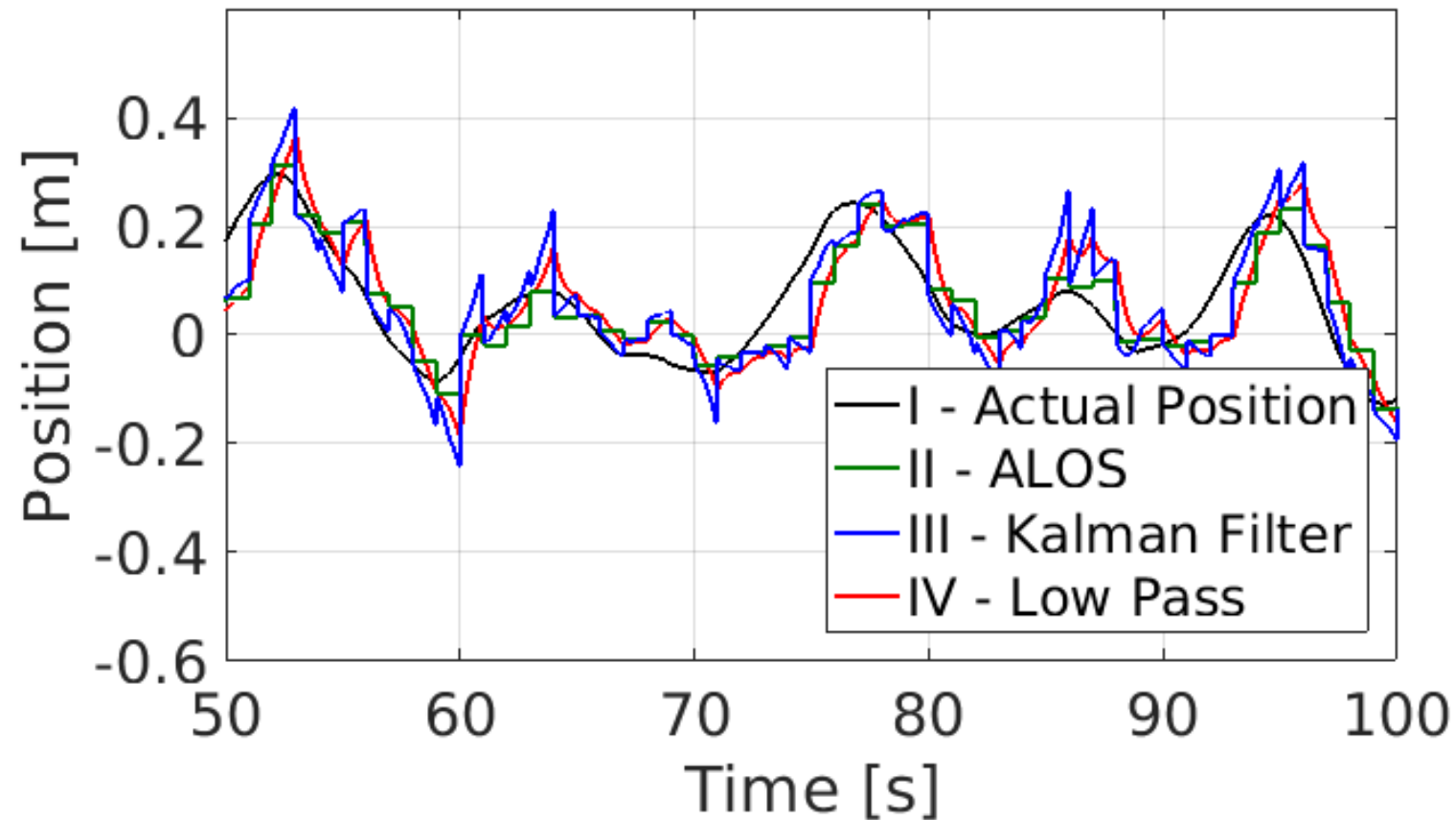
- ALOS box in is a mathematical approximation of the real system performance:
 - accuracy of 3 cm;
 - transport delay of 0.5 s; and
 - 1 Hz operation rate.



Results simulation trial

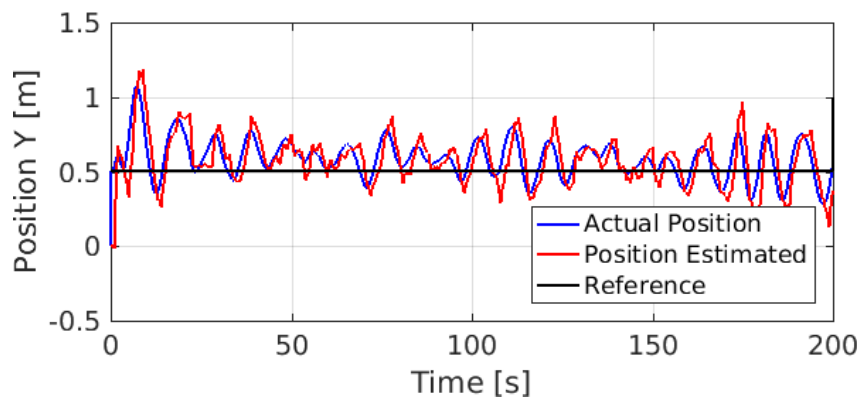
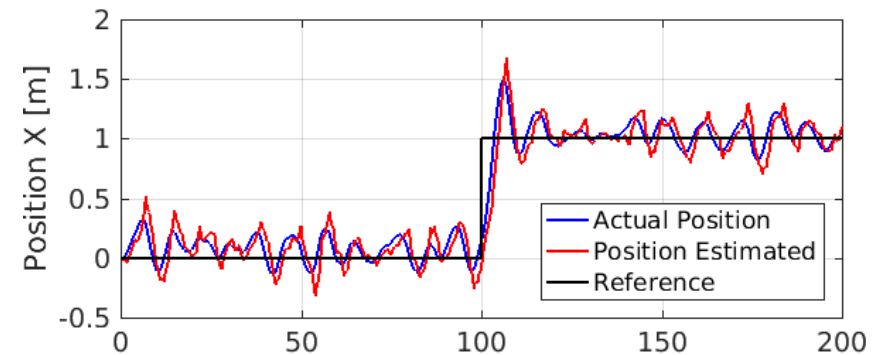
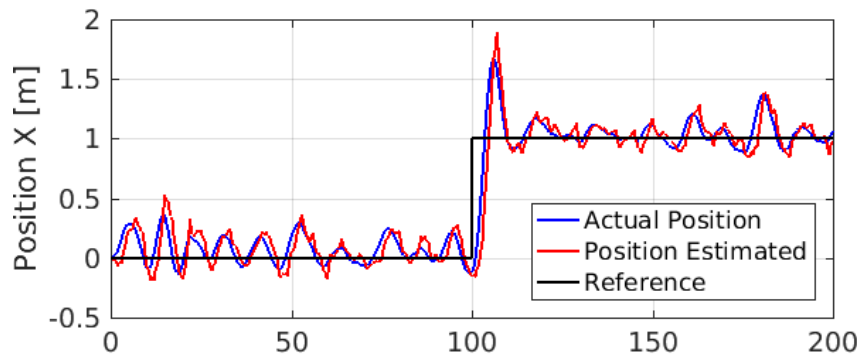
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Results simulation trial

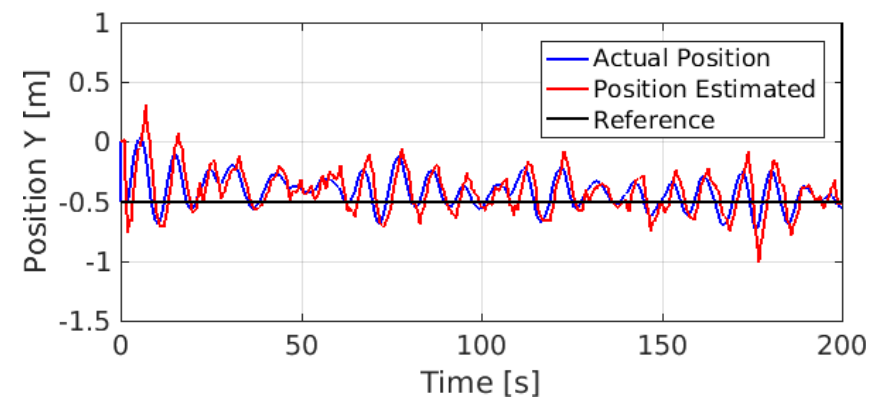


Results simulation trial

- Simultaneous change in the desired position



Quadrotor 1



Quadrotor 2

Results simulation trial

- In simulation, the error in the trajectory tracking task:
 - $X = 0.088$ m,
 - $Y = 0.094$ m,
 - $\psi = 1.882^\circ$

RMSE between output of the Kalman Filter and the real value



Conclusion

- Experimental results showed the centimeter level accuracy and other parameters of ALOS.
- Autonomous flight of quadrotors made using the complete architecture proposed (in simulation trial).
- Positive features of the system:
 - centimeter level accuracy of the position estimated;
 - real time operation capability;
 - use of low cost components for devices;
 - suitability for multiple vehicle scenarios.
- Future works:
 - use the complete system for real vehicles
 - also investigate the applicability in land robots